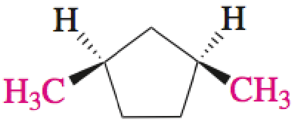
Chapter 4:
Chirality

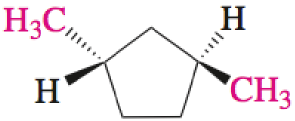
Additional Recommended Problems: Chp 4 – 28-31, 35-40

Levels of Organic Structures

- Organic compounds can be described using different *levels* of structure
  - **Composition:** molecular formula
  - **Constitution:** how atoms are connected; bonding patterns
  - **Configuration:** the three-dimensional positions of atoms
  - **Conformation:** positions of groups with respect to bond rotations and other “easy” processes
- Each level is associated with a type of isomerism
- **Configurational isomers** differ in configuration but are identical at higher levels



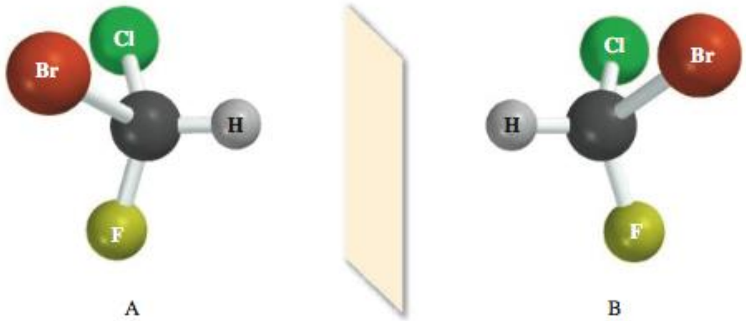
cis-1,3-Dimethylcyclopentane



trans-1,3-Dimethylcyclopentane

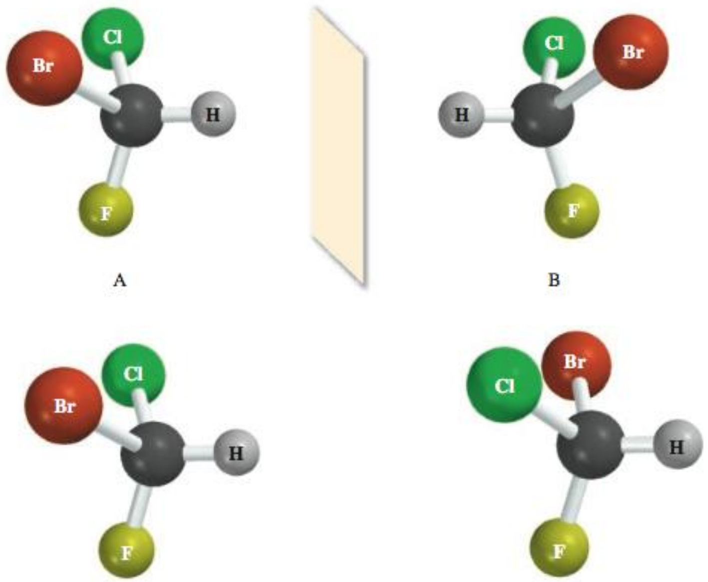
Chirality

- A **chiral** molecule is not identical to its mirror image, meaning it is not superimposable on its mirror image in three dimensions
- A molecule that is superimposable on its mirror image is called **achiral**
- Chiral mirror images exhibit an interesting kind of configurational isomerism



Chirality & Enantiomers

- Molecules A and B are mirror images, but they are not superimposable on one another!



It is impossible to align all five atoms in A and B.
A and B are both **chiral**.

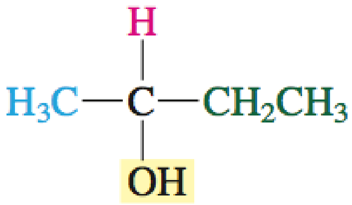
A and B are **enantiomers:** mirror images that are not identical.

The Chirality Center

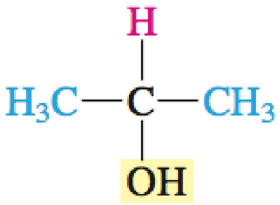
- A carbon bonded to four different groups is called a **chirality center**

- Designated by: (+) or (-), d or l, **(R) or (S)**

- Molecules with more than one chirality center may be achiral!



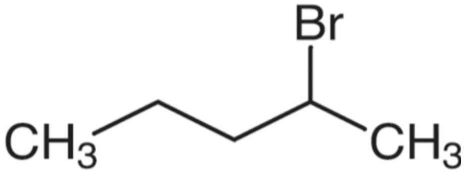
2-Butanol
Chiral; four different groups at C-2

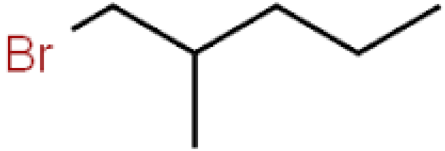


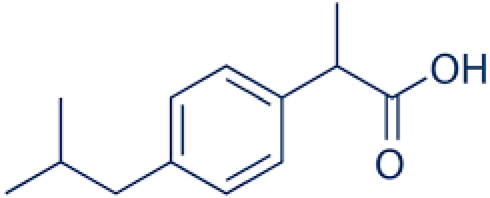
2-Propanol
Achiral; two of the groups at C-2 are the same

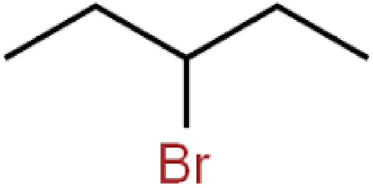
Chiral Center Identification

- Identify the chiral center(s), if any in each of the following molecules.



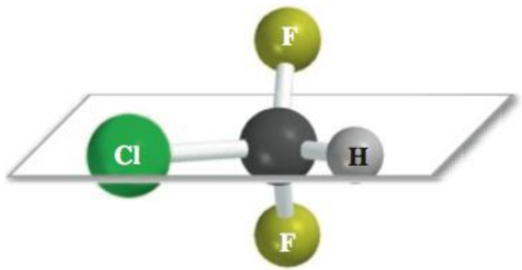


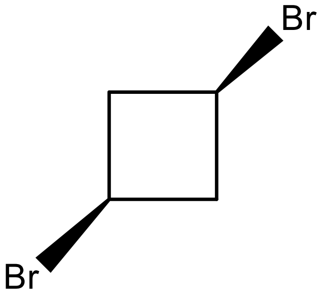




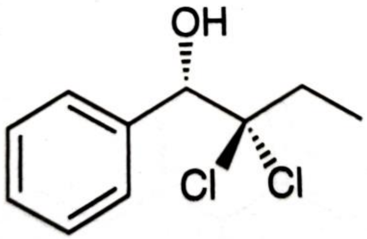
Shortcut for Identifying Chirality

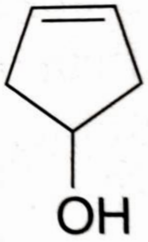
- Molecules that contain planes of symmetry (or inversion centers) are necessarily achiral
- Hence, we can look for a plane of symmetry to identify a molecule as achiral
- It is important to look *hard* when applying this method!
- Generating the mirror image and trying to superimpose it on the original is the *failsafe* method



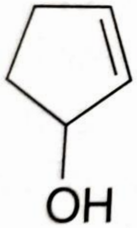


Identify Each Compound as Chiral or Achiral





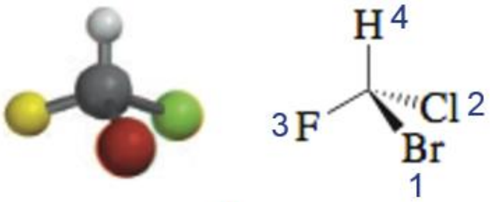
3-Cyclopentenol



2-Cyclopentenol

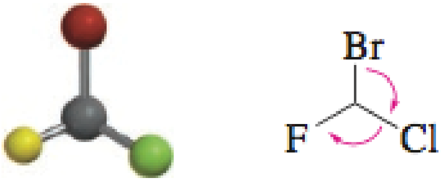
Cahn-Ingold-Prelog R,S System

1. Prioritize groups by atomic number, giving the higher atomic number priority (Br>Cl>F>H)



A

2. Point the group with lowest priorty (4) back.



A

3. Examine the direction of rotation from 1-3.

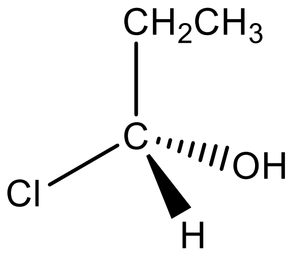
Clockwise = *R*; Counterclockwise = *S*

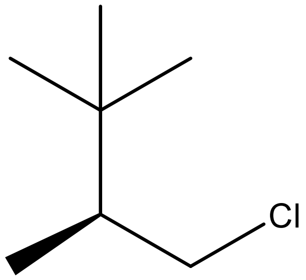
Prioritization Rules

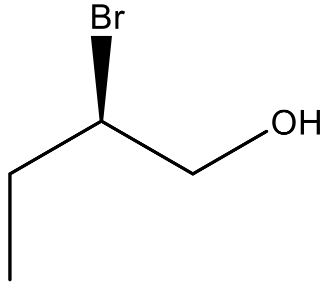
A REVIEW OF CAHN-INGOLD-PRELOG RULES: ASSIGNING THE CONFIGURATION OF A CHIRAL CENTER

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Identify the four atoms directly attached to the chiral center.	Assign a priority to each atom based on its atomic number. The highest atomic number receives priority 1, and the lowest atomic number (often a hydrogen atom) receives priority 4.	If two atoms have the same atomic number, move away from the chiral center looking for the first point of difference. When constructing lists to compare, remember that a double bond is treated as two separate single bonds.	Rotate the molecule so that the fourth priority is on a dash (going behind the plane of the page).	Determine whether the sequence 1-2-3 follows a clockwise order (*R*) or a counterclockwise order (*S*).

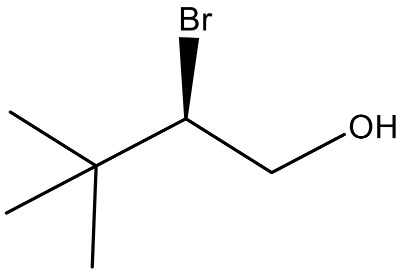
Assign absolute configurations as R or S to each of the following molecules.

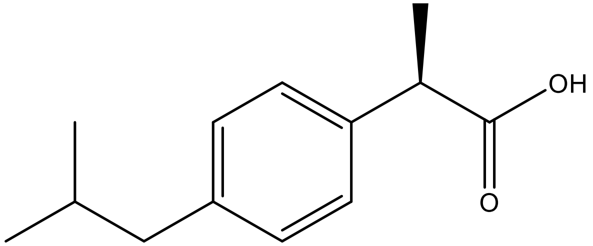


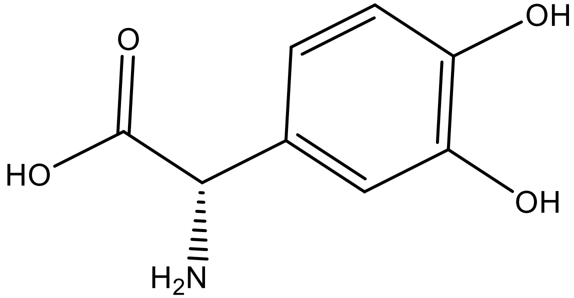


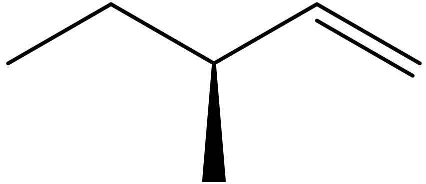


Assign the Configuration of the Chiral Center



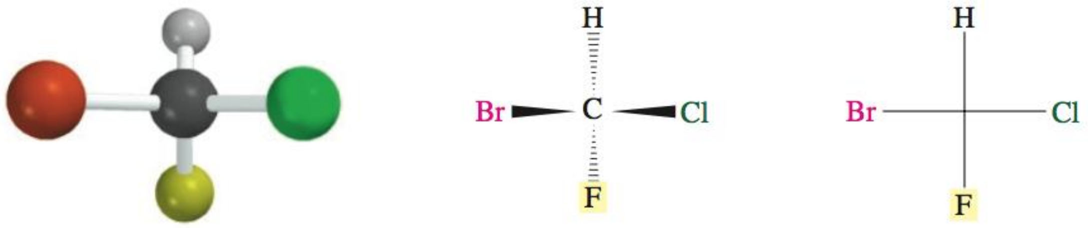




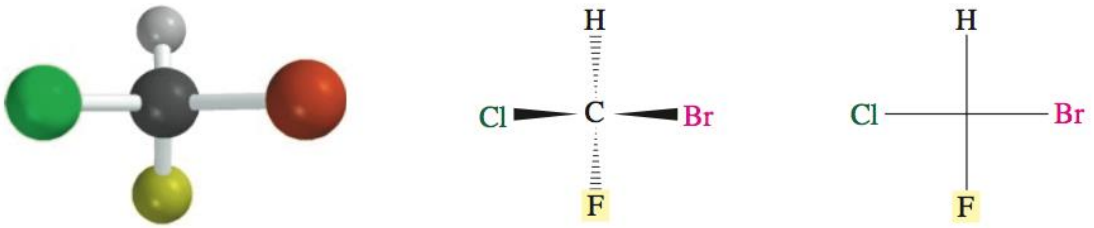


Draw (R)-2-chlorobutane and its enantiomer

Fischer Projections



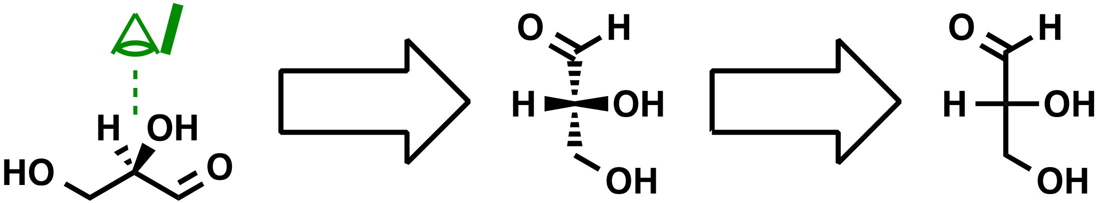
(*R*)-Bromochlorofluoromethane



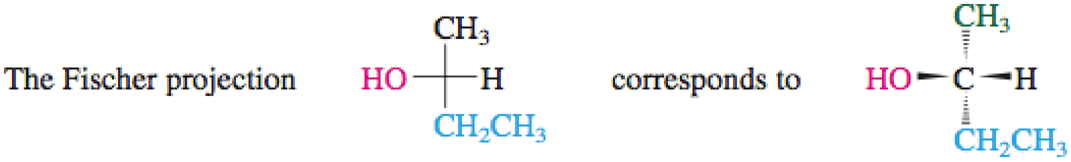
(*S*)-Bromochlorofluoromethane

Fisher Projections-1

- **Fischer projections** avoid the use of wedges and dashes by making use of a *standard orientation*, a "top-down" view of tetrahedral carbon



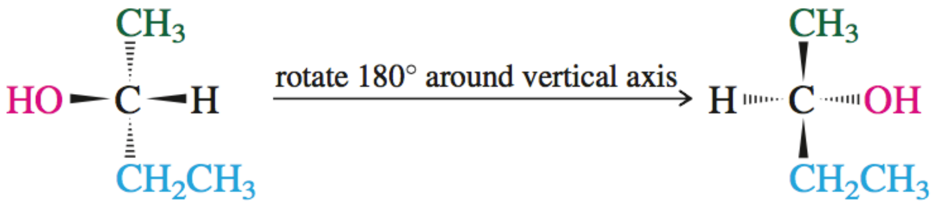
- Horizontal "arms" are pointing out to "hug" you; vertical bonds are pointing back away from you

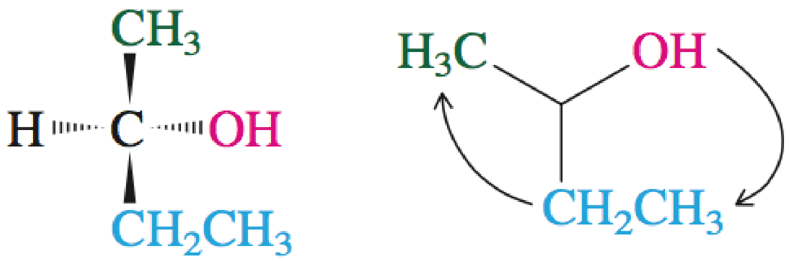


(*R*)-2-Butanol

R-*S* Descriptors and Fischer Projections

- To determine *R*/*S* for a stereocenter in a Fischer projection, rotate the projection so that the lowest priority group is *vertical* (pointed away from you)

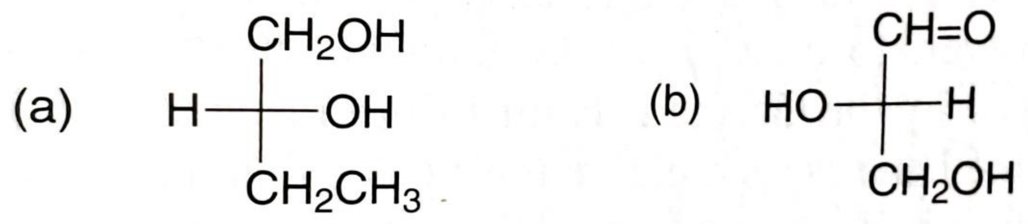




Clockwise rotation ➔ (*R*) configuration

Fischer Projection Practice Problem

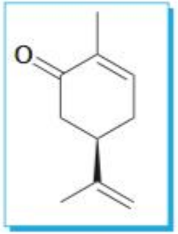
- Determine the absolute configuration represented by the Fischer projections



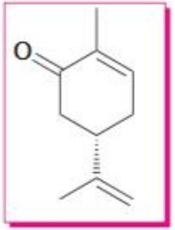
Enantiomers

- Enantiomeric molecules have identical sets of internal distances
- Thus, the physical properties of pure enantiomers are identical
  - Boiling point
  - Melting point
  - Density
- Enantiomers *differ* in their interactions with chiral molecules or phenomena!

Spearmint leaves



(*R*)-(−)-Carvone
(from spearmint oil)



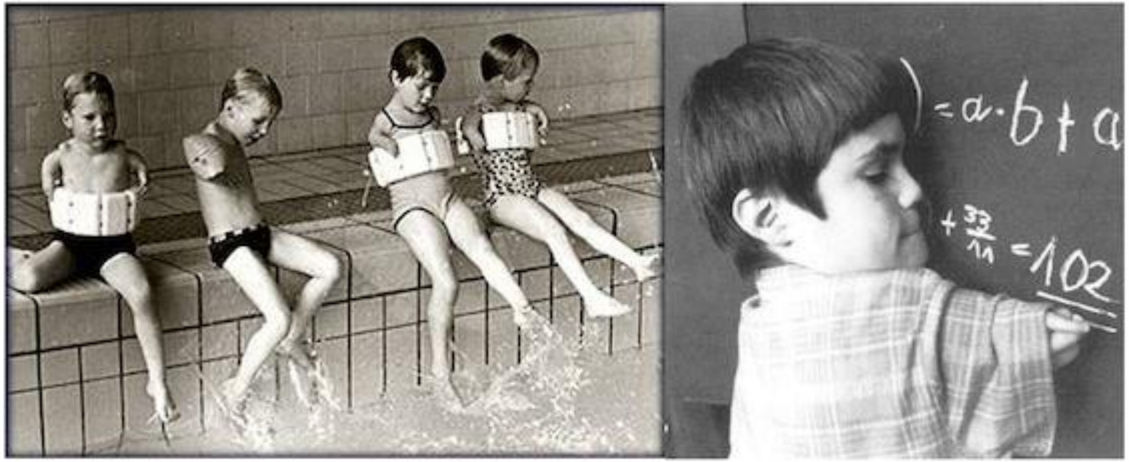
(*S*)-(+)-Carvone
(from caraway seed oil)

Caraway seeds

Olfactory receptors are chiral!

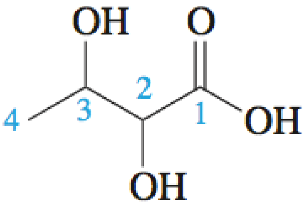
Why Care about Chirality?

- The miracle morning sickness drug, thalidomide (late 1950s)



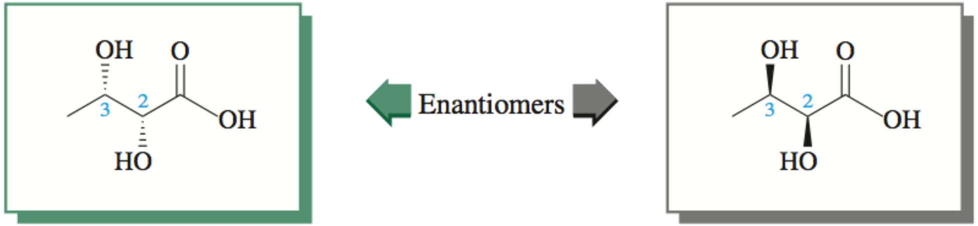
Chiral Molecules with Two Chirality Centers

- A chiral molecule with two stereocenters has *four* possible stereoisomers…
  - **I:** (2*R*,3*R*)
  - **II:** (2*S*,3*S*)
  - **III:** (2*R*,3*S*)
  - **IV:** (2*S*,3*R*)



2,3-Dihydroxybutanoic acid

- I and II are enantiomers, as are III and IV



III

(2*R*,3*S*) : $[\alpha]_D +17.8°$

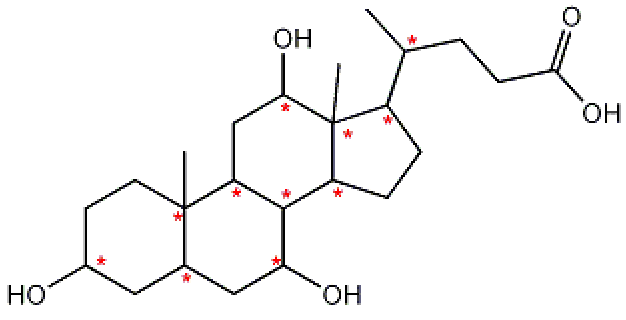
IV

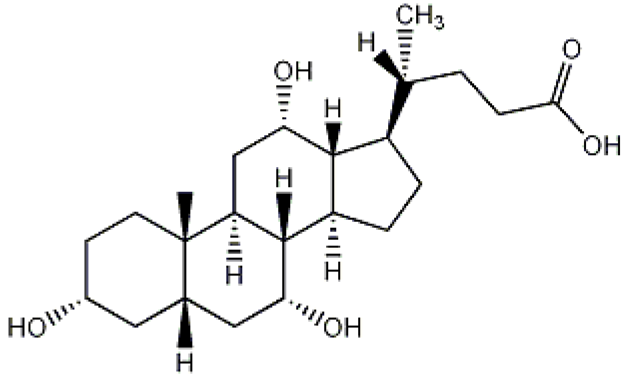
(2*S*,3*R*) : $[\alpha]_D -17.8°$

Are I and III related as enantiomers? What about II and IV?

2^n Rule

- Compounds with ***n*** stereocenters have a maximum of $\boldsymbol{2^n}$ stereoisomers

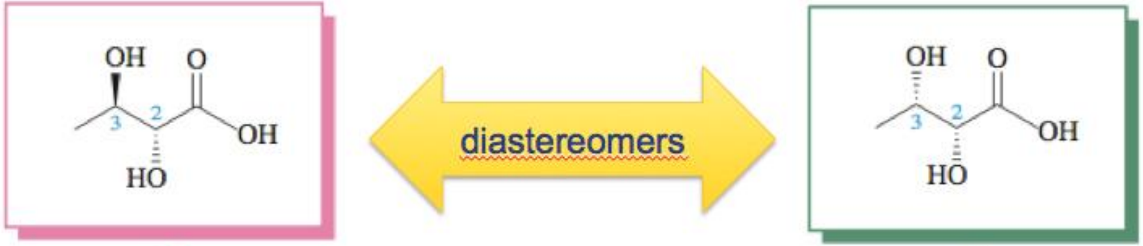




Natural cholic acid
1 of 2048 stereoisomers

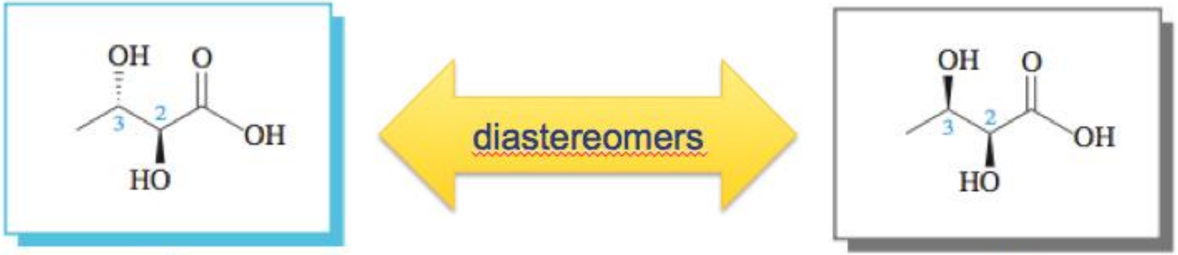
Diastereomers

- Isomers I and III differ at *some but not all* of their configurations. They are stereoisomers, but not mirror images (**diastereomers**)



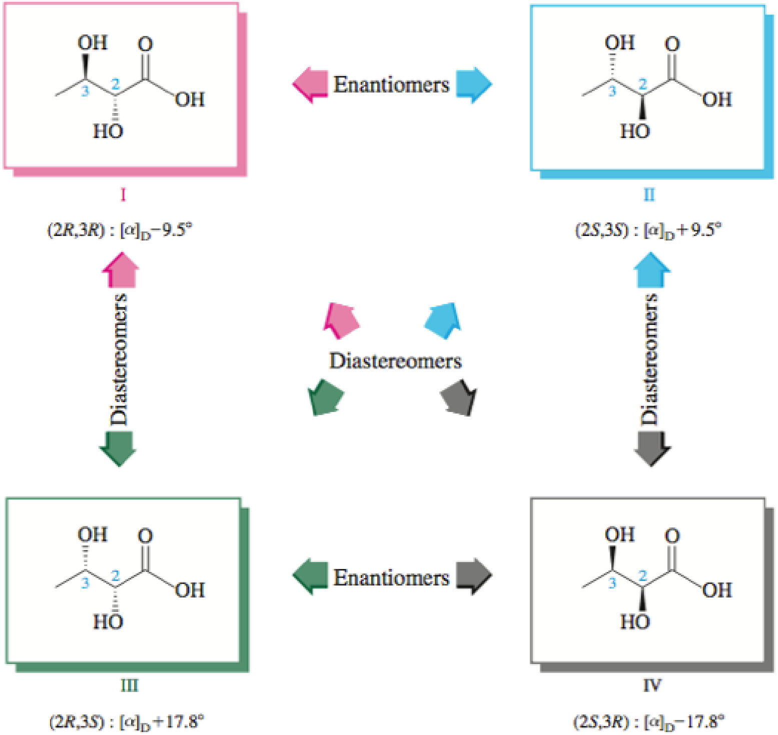
I III

- Isomers II and IV are also diastereomers



II IV

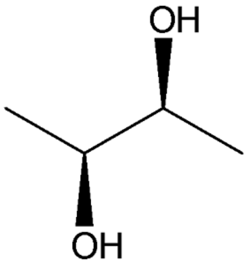
Isomers of 2,3-Dihydroxybutanoic Acid



Diastereomers have *different physical properties.*

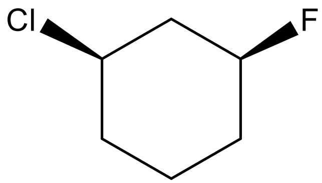
Meso Compounds

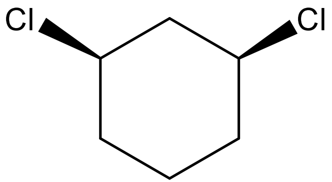
- Multiple stereocenters yet ***achiral***
- Mirror plane exists within molecule

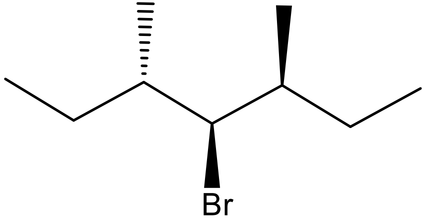


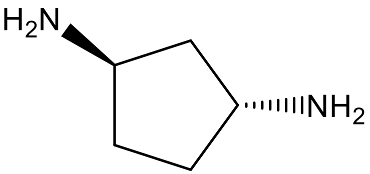
cis-1,2-dibromocyclobutane

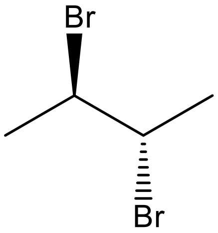
Which of the following are Meso?

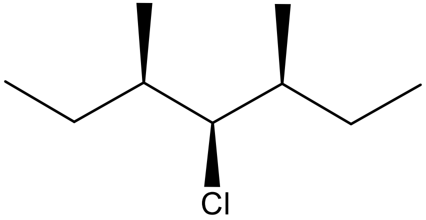




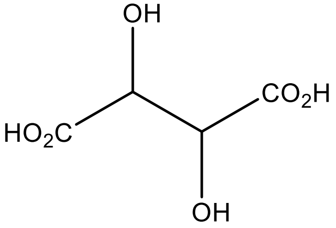


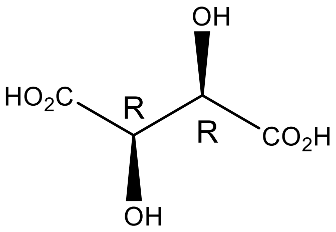




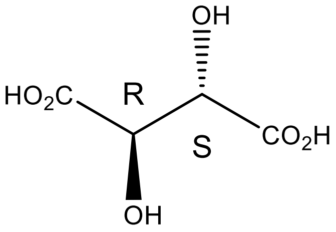


Tartaric Acid Meso Pairs





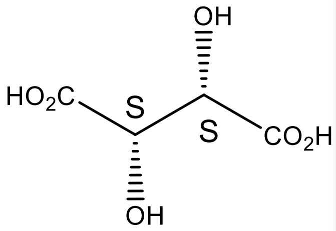
Diastereomers



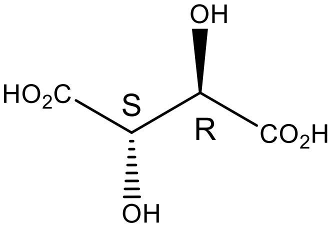
Enantiomers

Diastereomers

Meso pair = achiral
NOT enantiomers

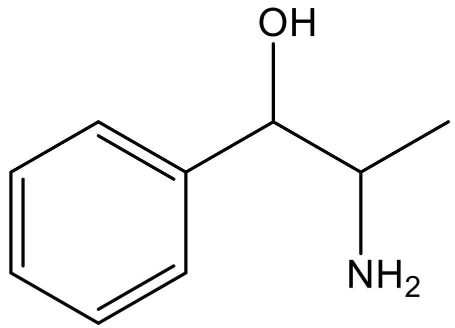


Diastereomers



Diastereomer/Enantiomer Example Problem

- Consider the molecule Ephedrine below. Draw all stereoisomers. Identify each as enantiomers or diastereomers to one another.



Identify the following structures as enantiomers, diastereomers, or meso